

INNOPOLIS UNIVERSITY · BACHELOR'S THESIS · 2026

# Secure Communication Protocols for Swarm Robotics

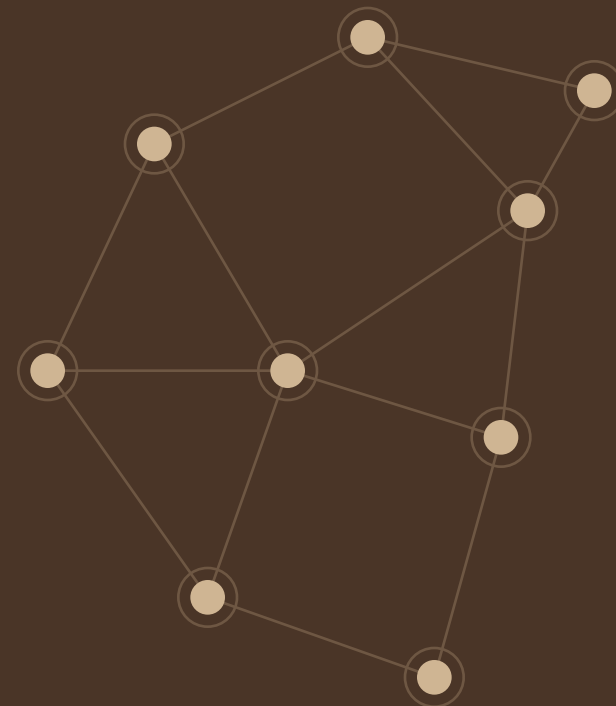
*An adaptive lightweight framework that switches cryptographic profiles at runtime  
— grounded in a reproducible benchmark.*

AUTHOR

**Georgii Butakov**

SUPERVISOR

**Oleg Bulichev**



## MOTIVATION

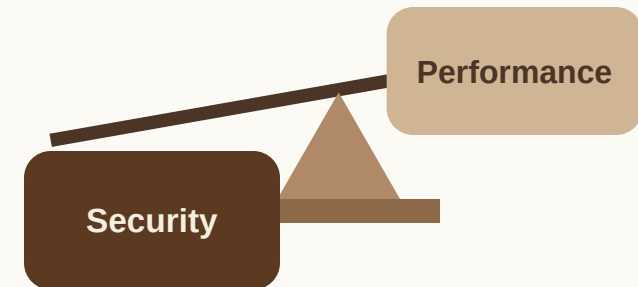
# One fixed cipher never fits the whole mission

Swarms coordinate through constant messaging over dynamic wireless links — on devices with limited energy and compute.

A permanently heavy profile wastes throughput and battery. A permanently **lightweight** one is unsafe when the threat level rises.

*Security and performance are coupled — they must be balanced at runtime, not fixed at design time.*

### THE CORE TRADE-OFF



*Adaptation re-balances  
as conditions change*

# Design it formally — then measure it

## GOAL

Design and evaluate a reproducible adaptive secure-communication framework for swarm-oriented systems that exposes the security–performance trade-off instead of assuming it.

01

### Formal model

Swarm as a dynamic graph with an adaptation function  $F : S(t) \rightarrow P$  mapping state to crypto profile.

02

### Working prototype

A pairwise benchmark with heavy, balanced, lightweight and adaptive operating modes.

03

### Bounded evidence

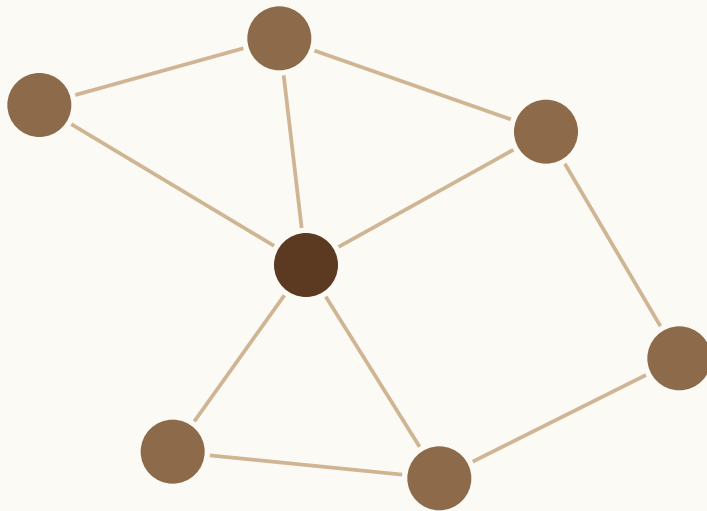
Repeatable measurements of latency, throughput, switching and a crypto-work metric under real constraints.

*Novelty: a bounded formal model paired with an executable benchmark — adaptive crypto is measured, not just proposed.*

# The swarm as a dynamic graph

$G ( V, E, t )$

nodes = agents · edges = live links



each node: position · energy · keys · CPU

## Adaptation function

STATE  $S(t)$

energy  $\bar{e}$

threat  $\theta$

density  $\rho$

packet loss  $\lambda$

PROFILE  $P$

heavy

balanced

lightweight

**F**

select

$F : S(t) \rightarrow P$  maps the observed runtime state to a discrete cryptographic profile — the optimisation view is approximated by a threshold selector.

# What we defend against — and guarantee

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## ADVERSARY CAPABILITIES

**1** **Passive**  
eavesdrops on traffic

**2** **Active**  
modifies / injects messages

**3** **Compromised node**  
leaks keys & memory

## DATA-PLANE GUARANTEES

✓ **Confidentiality**  
AEAD encryption, unique nonce per message

✓ **Authentication**  
HMAC-SHA256 / sequence-bound hash token

✓ **Integrity**  
authenticated encryption tag

✓ **Compromise containment**  
per-epoch X25519 rekeying

*Scope is bounded to the data plane — a fully authenticated control plane is named as future work.*

# A hybrid stack — each primitive where it fits

## SESSION SETUP

**X25519 + HKDF-SHA256**

Ephemeral key exchange per epoch → forward secrecy

periodic

## PAYLOAD PROTECTION

**AES-GCM / ChaCha20-Poly1305**

AEAD confidentiality + integrity, 96-bit nonce, 128-bit tag

per message

## RUNTIME AUTH

**HMAC-SHA256 / hash token**

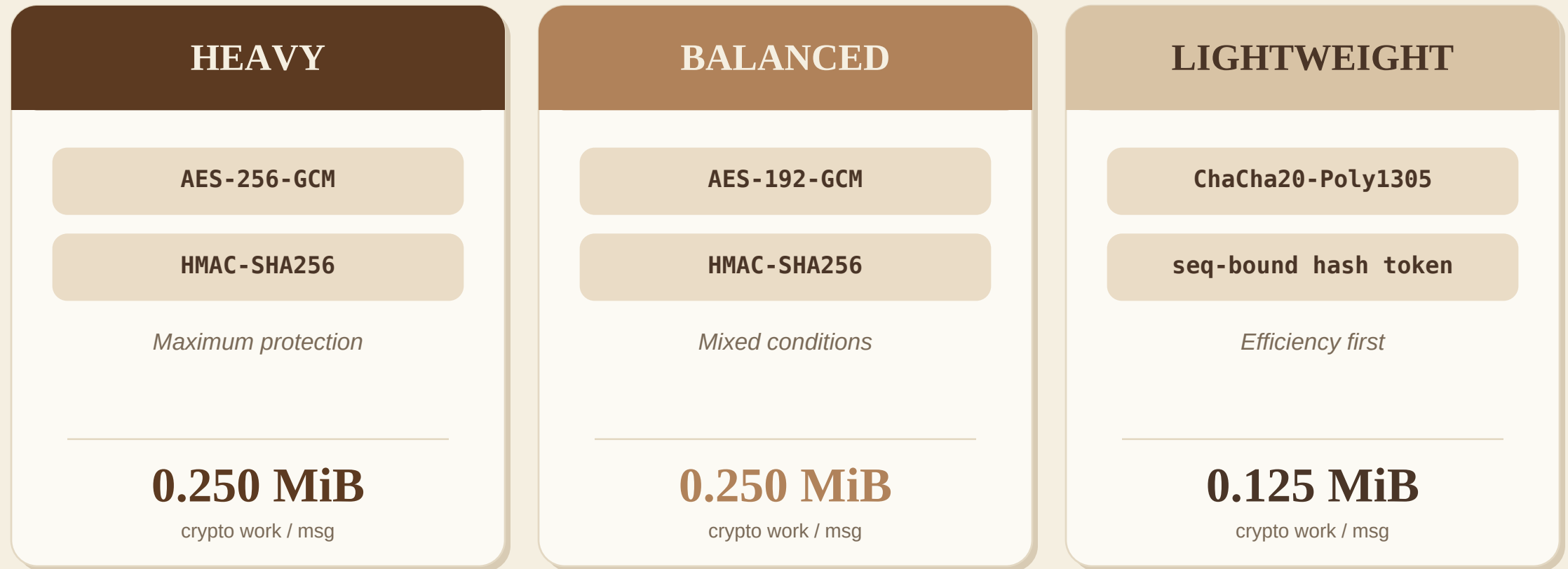
Lightweight origin & replay check on the data path

per message

## WHY

Expensive asymmetric work stays off the per-message path; only fast symmetric primitives run per packet.

## Three discrete operating points



*Lightweight does half the per-message crypto work of the authenticated profiles.*

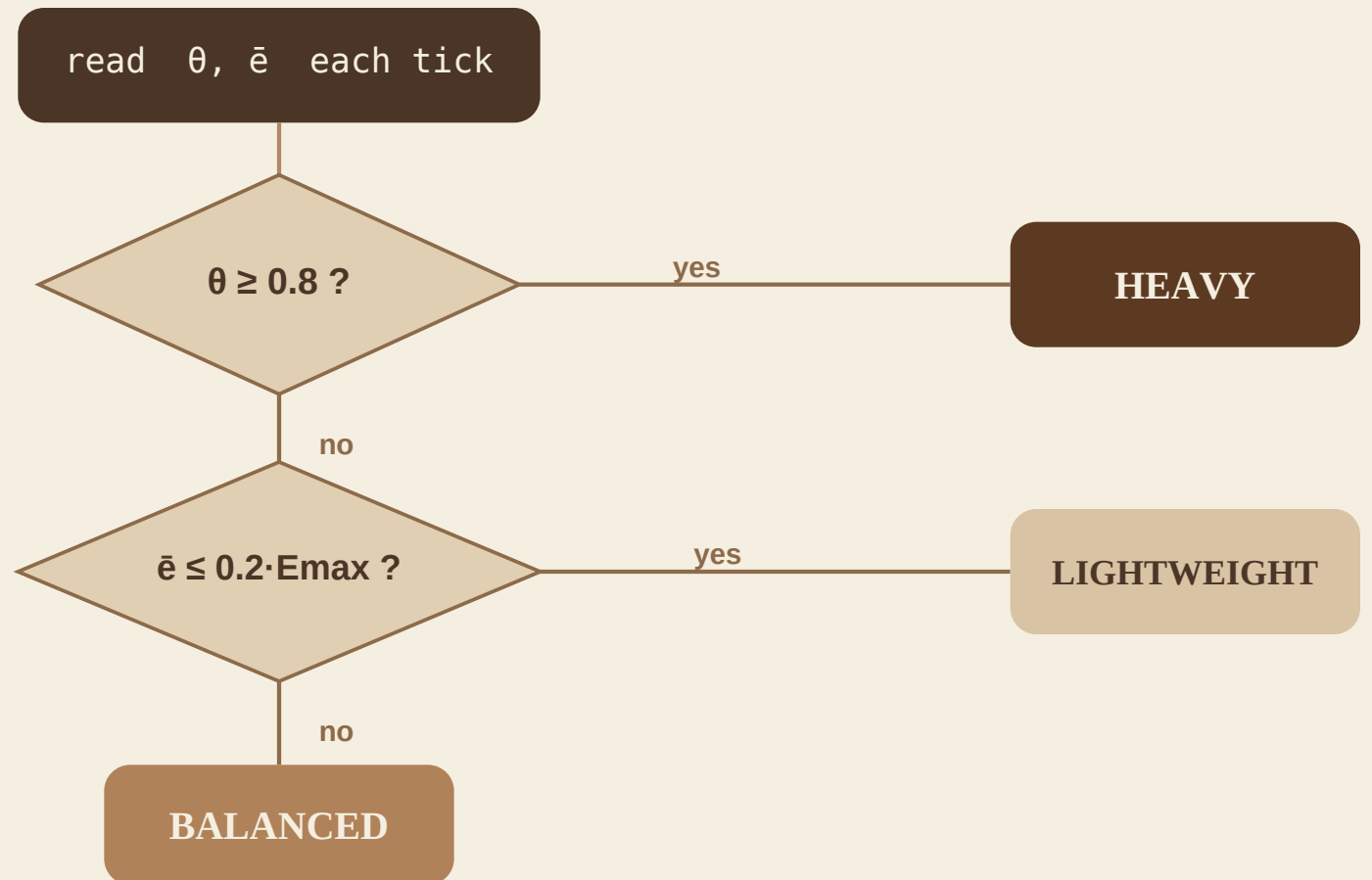
# A transparent threshold selector

## Two triggers

Escalate to heavy when the threat estimate is high.

Fall back to lightweight when energy runs low.

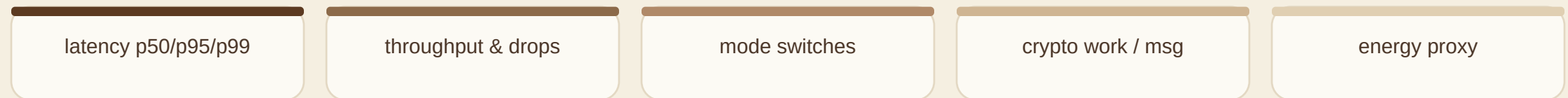
Otherwise stay balanced. The selector cost is negligible vs. packet protection.



# A constrained, reproducible two-peer testbed



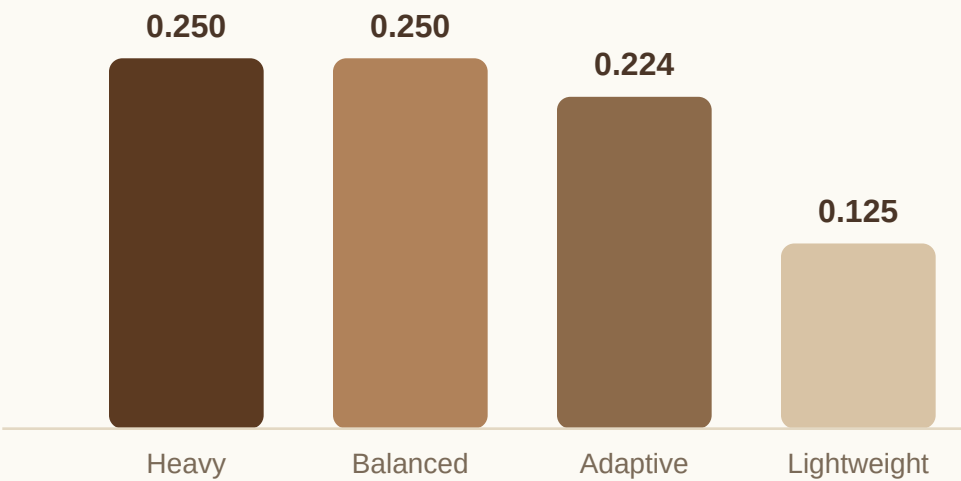
MEASURED PER RUN (60 s · 3 repeats · 228 summary rows)



# Adaptive spends protection where it is needed

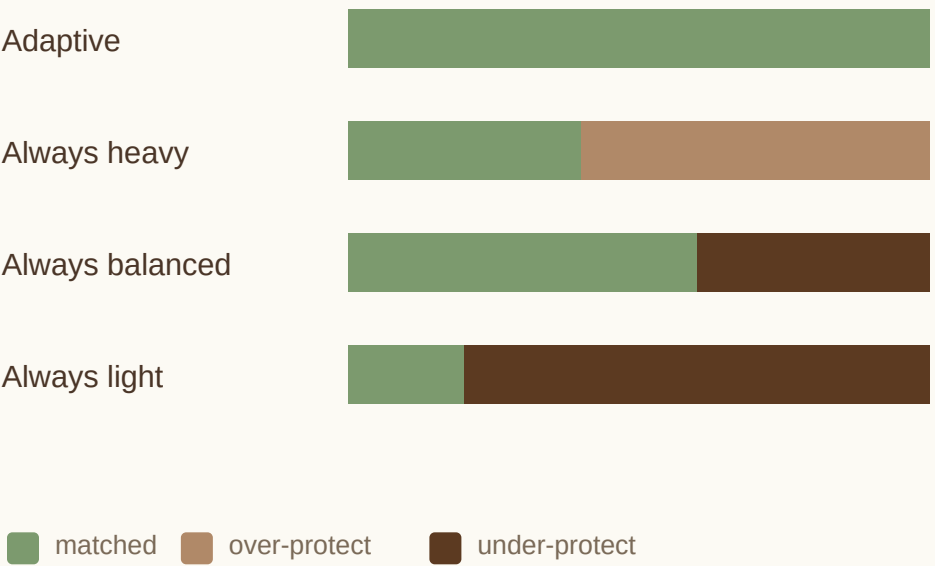
## Crypto work per delivered message

MiB · lower is cheaper



## Policy fit over the 60-s schedule

share of message-time at the right protection level

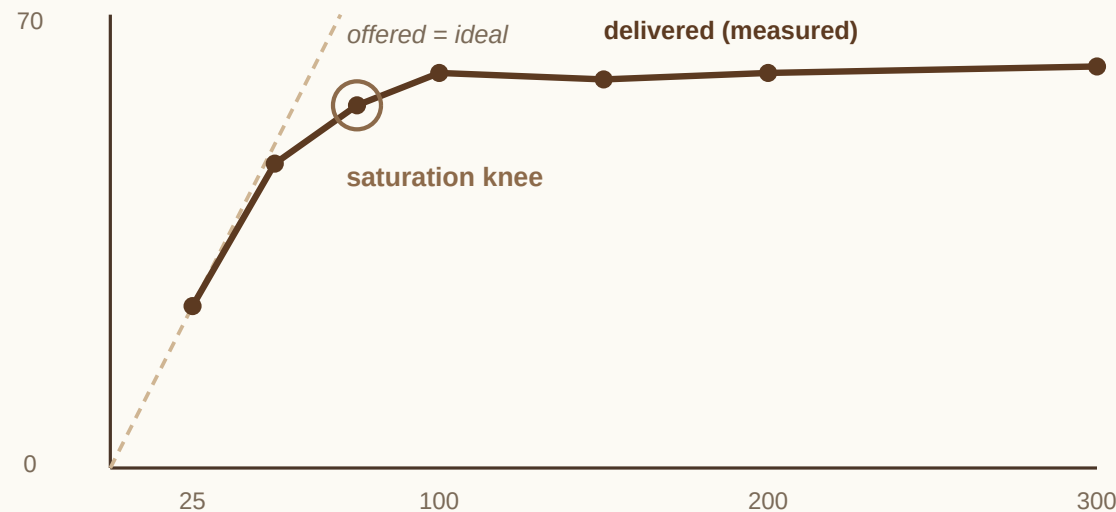


Adaptive matches the required profile for the whole schedule and saves 26.7% of protection cost vs. always-heavy — without the under-protection of always-light.

## Under a tight CPU cap, transport dominates

### Delivered throughput vs. offered rate

0.05-core cap · msg/s



### CPU CAP REMOVES THE BOTTLENECK

0.05 core ██████████ 65% drop

0.10 core ██████████ 22% drop

0.25 core ██████████ no drop

The knee is CPU-bound, not a crypto-design limit: at 0.25 core all four modes sustain the full rate.

Above ~75 msg/s the prototype is queue-limited, so end-to-end curves converge — the profile difference shows up in crypto work, not wall-clock latency.

## CONCLUSION

# Adaptive crypto, measured — not just proposed

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### Formalised

swarm graph + adaptation function  $F : S(t) \rightarrow P$



### Built

pairwise prototype with heavy / balanced / lightweight / adaptive



### Measured

lightweight is the cheapest default; adaptive allocates protection correctly over time

## WHAT COMES NEXT

### 1 Authenticated control plane

signed bootstrap & profile updates

### 2 Multi-peer validation

churn, contention, async switching

### 3 Hardware & post-quantum

real platforms; hybrid PQ session setup

*Thank you — questions welcome.*

[github.com/mazzzz3r/adaptive-swarm-crypto-thesis](https://github.com/mazzzz3r/adaptive-swarm-crypto-thesis)